

Daniel Ari Friedman, PhD

Computational biologist, Active Inference researcher, cognitive security researcher, educator, and artist | California, USA, Earth

Verify	resume/verify.html QR target, public hashes, source files, and artifact sizes.
Source	manifest sha256 9f491e932e9213d1 resume/source.json + canonical data exports
Output	json sha256 1d5393d78d53ca6f data/resume.json resume/resume.pdf



Curated Works 184 data/works.json	Software Rows 95 data/software.json	Scholar Metrics 777 Scholar profile	Resume Records 119 verify page
Source Hash 9f491e932e92 source manifest	JSON Hash 1d5393d78d53 data/resume.json	PDF Hash verify page resume/verify.html	Footer QR every page resume/verify.html

Source-to-Output Flow

Inputs	Generated JSON	Public PDF + Verification
resume/source.json data/works.json data/software.json source sha256 9f491e932e9213d1	data/resume.json 209,917 bytes json sha256 1d5393d78d53ca6f	resume/resume.pdf resume/verify.html PDF hash posted on verification page

Section Counts

Education	4
Experience	16
Awards	12
Conferences	17
Media / Outreach	29
Service	27
Art Uses	14



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GitHub https://github.com/docxology	LinkedIn https://www.linkedin.com/in/danielarifriedman/
Scholar https://scholar.google.com/citations?user=DXjPFtYAAAAJ	ORCID 0000-0001-6232-9096
Keybase / ENS docxology / docxology.eth	Art https://www.flickr.com/photos/daniel_friedman/

Curated Works 184 from data/works.json	Software Rows 95 61 owned + 34 All	Scholar Citations 777 h 15 / i10 17 as of 2026-06-09
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Variant: full. Generated: 2026-07-07T00:59:17Z. Current public metrics: 184 curated works; 95 catalogued software repositories (61 owned + 34 All); 777 Google Scholar citations, h-index 15, i10-index 17 as of 2026-06-09.

Summary

Multidisciplinary researcher and open-source builder working across social insect biology, Active Inference, cognitive security, computational frameworks, education, and art.

Education (4)

2019-2023 Postdoctoral Fellow | University of California, Davis | Davis, CA, USA

- Co-mentors: Prof. Brian Johnson (UC Davis) and Prof. Tim Linksvayer (Arizona State University).
- NSF Postdoctoral Research Fellowship in Biology (PRFB), award DBI-2010290; NSF budget period 2020-2022 with affiliation extending to 2023.
- Genomics, epigenomics, transcriptomics, and behavior in eusocial insects, mainly ants and bees.
- Science research, mentoring, participation, service, and organizing events and communities online.

Source: NSF DBI-2010290 <https://grantome.com/grant/NSF/DBI-2010290> | UC Davis <https://www.ucdavis.edu/>

2014-2019 PhD | Stanford University | Stanford, CA, USA

- Advisor: Professor Deborah M. Gordon, Biology.
- Research, teaching, service, and coursework from Summer 2014 to Summer 2019.
- Thesis topics: ant ecology, collective behavior, brain gene expression, and neurometabolism.
- Stanford Graduate Fellow (Urbanek Family Fellowship), 2014-2017.
- Co-organizer of the Stanford Complexity Group, 2016-2019.
- Stanford Neuroscience NeuroChoice Grant for neurometabolism in ants, 2019, \$25,000.
- Stanford Systems Biology Seed Grant for gene expression and behavior in ants, 2018, \$25,000.
- Theodore Roosevelt Memorial Award, American Museum of Natural History, 2016, \$2,500.
- Lewis and Clark Fund grant, American Philosophical Society, 2016, \$4,000.
- NSF GRFP Honorable Mention, 2015 and 2016.

Source: Stanford dissertation <http://purl.stanford.edu/pb813wm1484>



2010-2014 BSc in Genetics | University of California, Davis | Davis, CA, USA

- BSc in Genetics, University of California, Davis.
- Genetics Department Citation, Highest Honors, 2014; 3.74 GPA overall.
- Campus Community Service Silver Award, 2014.
- Stanley and Jacqueline Schilling Award for Service, 2014.
- Completed the Population Biology Graduate Group one-year base PhD curriculum, 2013.
- Co-founder of The Aggie Transcript, an undergraduate biology journal, 2014.
- Victoria Finnerty Award for Research in Drosophila, 2013.
- Elected member of Phi Beta Kappa, UC Davis Chapter, 2014.
- Co-organizer of the Chess Club at UC Davis, 2012-2014.
- Research Scholars Program in Insect Biology, 2011-2014.
- Provost's Undergraduate Fellowship for Drosophila research, 2011, \$2,500.

Source: UC Davis <https://www.ucdavis.edu/>

2006-2010 High School | Crystal Springs Uplands School | Hillsborough, CA, USA

- Crystal Springs Uplands School.

Source: Crystal Springs Uplands School <https://www.crystal.org/>

Experience (16)**2026-ongoing Del Norte Triplicate | Typographical Editor**

Community Organization

- Typographical editing for The Triplicate newspaper of Del Norte County and Crescent City.

Source: The Triplicate <https://www.triplicate.com/>

2021-ongoing Active Inference Institute | Co-founder, Participant, Officer, President, Treasurer

Community Organization

- Co-founded the Active Inference Institute; current President and Treasurer.

Source: All officers <https://www.activeinference.institute/officers> | All <https://activeinference.org/>

2019-2024 Complexity Adventures | Co-founder, Facilitator, Participant, Organizer

Community Organization

- Organized activities and community around learning and applying complexity science.

Source: Complexity Adventures profile <https://www.complexityadventures.com/organizers/2>

2019-ongoing COGSEC | Researcher

Community Organization

- Research facilitation and participation on initiatives related to cognitive security.

Source: COGSEC <https://www.cogsec.org/>

2014-2019 Stanford Complexity Group | Organizer, Participant

Community Organization

- Organized the Stanford Complexity Group while it was active.

Source: Stanford <https://www.stanford.edu/> | verification <https://danielarifriedman.com/resume/verify.html>

2025-2026 Stealth Startup | Operator

Startup

- Systems modeling and related work to be public later.

Source: verification <https://danielarifriedman.com/resume/verify.html>

2023-2024 Stealth Startup | Co-founder, Operator

Startup

- Stealth Active Inference startup.

Source: verification <https://danielarifriedman.com/resume/verify.html>



2019-2023 Remotor | Co-founder, Researcher

Startup

- Education, online event planning, and consulting for remote teams.

Source: verification <https://danielarifriedman.com/resume/verify.html>

2017-2019 fm.analytics | Co-founder, Researcher

Startup

- Predictive analytics and dashboard visualizations with Prof. Glenn Magerman at KU Leuven.

Source: verification <https://danielarifriedman.com/resume/verify.html>

2017-2019 Cryptoptera | Co-founder, Researcher

Startup

- Three-person cryptocurrency research group.

Source: verification <https://danielarifriedman.com/resume/verify.html>

2019-2023 Johnson Lab / Linksvayer Lab | Mentor, Researcher

Research

- Postdoctoral position during NSF fellowship with Brian Johnson at UC Davis and Tim Linksvayer at Arizona State University.

Source: Johnson Lab <https://johnsonlab.ucdavis.edu/> | UC Davis <https://www.ucdavis.edu/>

2014-2019 Gordon Lab | Mentor, Researcher

Research

- Graduate research, field work, wet lab, bioinformatics, and mentoring of high schoolers, undergraduates, graduate students, and non-academics.

Source: Gordon Lab <https://web.stanford.edu/~dmgordon/>

2010-2014 Kopp Lab | Researcher

Research

- Undergraduate research assistant with Prof. Artyom Kopp in Ecology and Evolution at UC Davis, studying non-coding DNA elements in the evolution of the Drosophila foreleg sex comb.

Source: UC Davis Biology <https://biology.ucdavis.edu/>

2014-2019 Pedicab People Movers | Operator

Pedicab

- Operated a pedicab/rickshaw at events including music festivals in California and downtown areas in Davis and San Jose.

Source: verification <https://danielarifriedman.com/resume/verify.html>

2015-2016 Stanford University | Teacher, Organizer, Graduate Teaching Assistant

Teaching

- Graduate Teaching Assistant for undergraduate and graduate courses: General Biology: Cell and Molecular Biology (2015) and Ethics of Evolutionary Biology (2016).

Source: Stanford Biology <https://biologyv.stanford.edu/>

2014 UC Davis | Teacher, Organizer, Course Co-developer

Teaching

- Student teacher and developmental biology course co-developer with Prof. Susan Keen at UC Davis.

Source: UC Davis Biology <https://biology.ucdavis.edu/>

Awards and Fellowships (12)

Year	Award	Details
2020	NSF Postdoctoral Research Fellowship in Biology	Award DBI-2010290; NSF budget period 2020-2022.
	Source: NSF DBI-2010290 https://grantome.com/grant/NSF/DBI-2010290	



Year	Award	Details
2014	Stanford Graduate Fellow, Urbanek Family Fellowship Source: Stanford Graduate Fellowship https://vpqe.stanford.edu/fellowships-funding/sqf	Stanford University, 2014-2017.
2019	Stanford Neuroscience NeuroChoice Grant Source: Stanford Neurosciences Institute https://neuroscience.stanford.edu/	Neurometabolism in ants, \$25,000.
2018	Stanford Systems Biology Seed Grant Source: Stanford Systems Biology https://sysbio.stanford.edu/	Gene expression and behavior in ants, \$25,000.
2016	Theodore Roosevelt Memorial Award Source: AMNH Theodore Roosevelt Grants https://www.amnh.org/research/richard-gilder-graduate-school/academics-and-research/fellowships-and-grants/theodore-roosevelt-memorial-grants	American Museum of Natural History, \$2,500.
2016	Lewis and Clark Fund grant Source: APS Lewis and Clark Fund https://www.amp-hilsoc.org/grants/lewis-and-clark-fund-exploration-and-field-research	American Philosophical Society, \$4,000.
2015	NSF GRFP Honorable Mention Source: NSF GRFP https://www.nsfgrfp.org/	Honorable Mention in 2015 and 2016.
2014	UC Davis Genetics Department Citation Source: UC Davis Genetics https://genetics.ucdavis.edu/	Highest Honors.
2014	UC Davis service awards Source: UC Davis https://www.ucdavis.edu/	Campus Community Service Silver Award and Stanley and Jacqueline Schilling Award for Service.
2013	Victoria Finnerty Award for Research in Drosophila Source: UC Davis Genetics https://genetics.ucdavis.edu/	UC Davis.
2014	Phi Beta Kappa Source: Phi Beta Kappa https://www.pbk.org/	Elected member, UC Davis Chapter.
2011	Provost's Undergraduate Fellowship Source: UC Davis Undergraduate Research Center https://urc.ucdavis.edu/puf	Drosophila research, \$2,500.



Works and Publications (184)

#	Year	Type	Title and Venue	Link
001	2026	Paper	A template/ approach to Reproducible Generative Research: Architecture and Ergonomics from Configuration through Publication Zenodo	10.5281/zenodo.16903351
002	2026	Series	Active Inference Journal — 500+ videos on Active Inference with transcripts YouTube	https://video.activeinference.institute/
003	2026	Paper	Before Pragmatism Had a Name: Blake's "America A Prophecy" Anticipates American Anticipatory Epistemology Zenodo	10.5281/zenodo.18807970
004	2026	Paper	The Golden Compass and the Lunar Flux: William Blake, Bimetallic Meta-stability, and the Architecture of Value Zenodo	10.5281/zenodo.19335195
005	2026	Paper	Cognitive Integrity Framework: Formal Foundations for Multiagent Security (Part 1: Theory) Zenodo	10.5281/zenodo.18364118
006	2026	Series	Personal YouTube — 200+ livestreams: drawings, Synergetics, paper discussions YouTube	https://www.youtube.com/@danielarifriedman/playlists
007	2026	Paper	The Doors of Perception are the Threshold of Prediction: Active Inference and William Blake's Theory of Seeing Zenodo	10.5281/zenodo.18600040
008	2025	Presentation	5th Applied Active Inference Symposium — Abstract Book Presentation	10.5281/zenodo.17555266
009	2025	Paper	Adaptive Basic Income (AuBI): Integrating AI, Decentralized Infrastructure, and Active Inference Zenodo	10.5281/zenodo.17228945
010	2025	Paper	CEREBRUM: Case-Enabled Reasoning Engine with Bayesian Representations for Unified Modeling Zenodo	10.5281/zenodo.15170907
011	2025	Paper	Computational Complexity and Energetics of the Ant Stack Zenodo	10.5281/zenodo.17238736
012	2025	Paper	EvoJump: Stochastic Modeling of Evolutionary Ontogenetic Trajectories Zenodo	10.5281/zenodo.17229924
013	2025	Paper	Increasing the Accessibility and Applicability of Active Inference Zenodo	10.5281/zenodo.15061666
014	2025	Paper	MDKV: A Multitrack Markdown Container for Structured, Portable Documents Zenodo	10.5281/zenodo.16790554
015	2025	Paper	Markdown Decision Process: A Framework for Probabilistic Document Analysis Zenodo	10.5281/zenodo.17244386
016	2025	Paper	On Cognitive Art & Science: Toward Wholeness From Both Sides Zenodo	10.5281/zenodo.16740438
017	2025	Paper	On Time Zenodo	10.5281/zenodo.15168381
018	2025	Paper	QuadMath: An Analytical Review of 4D and Quadray Coordinates Zenodo	10.5281/zenodo.16887799
019	2025	Paper	ResNei: Solution Design Document Zenodo	10.5281/zenodo.15389682
020	2025	Paper	Symergetics: Symbolic Synergetics for Rational Arithmetic Zenodo	10.5281/zenodo.17114389
021	2025	Paper	Synthesis of Agent and Niche Zenodo	10.5281/zenodo.17235137
022	2025	Presentation	Systems Processes, Active Inference, and Beyond Presentation	10.5281/zenodo.17138223
023	2025	Paper	Temporal Depth in a Coherent Self and in Depersonalization Frontiers in Psychology	10.3389/fpsyg.2025.1585315
024	2025	Paper	The Active Inference Institute & Active Inference Ecosystem (v3, 2025 snapshot) Zenodo	10.5281/zenodo.17982447
025	2025	Paper	The Ant Stack Zenodo	10.5281/zenodo.16782756
026	2025	Paper	The Discovery Engine: AI-Driven Synthesis and Navigation of Scientific Knowledge Landscapes ArXiv	10.48550/arXiv.2505.17500
027	2025	Paper	Thoughtseeds: A Hierarchical and Agentic Framework for Investigating Thought Dynamics in Meditative States Entropy	10.3390/e27050459
028	2025	Paper	Towards a Science of Consciousness and Social Complexity... For Ants Book Chapter	https://www.editorafi.org/ebook/c128-colonias-formigas-conscientes
029	2024	Paper	Aligning Active Inference Ontology to SUMO Zenodo	10.5281/zenodo.11459322



#	Year	Type	Title and Venue	Link
030	2024	Presentation	BioFirm Development at Applied Active Inference Symposium 2024 Presentation	10.5281/zenodo.14861595
031	2024	Paper	Bridging gaps in image meme research: A multidisciplinary paradigm JASIST	10.1002/asi.24900
032	2024	Paper	Chemical and transcriptomic diversity do not correlate with ascending levels of social complexity in Blattodea Ecology & Evolution	10.1002/ece3.70063
033	2024	Paper	Comments on National Digital Twins R&D Strategic Plan Zenodo	10.5281/zenodo.13273681
034	2024	Paper	Enhancing Population-based Search with Active Inference ArXiv	10.48550/arXiv.2408.09548
035	2024	Paper	FarmWorks: Decentralized AI Agents for Personalized Solutions Zenodo	10.5281/zenodo.13754585
036	2024	Paper	Federated inference and belief sharing Neuroscience & Biobehavioral Reviews	10.1016/j.neubiorev.2023.105500
037	2024	Paper	Four-fold Fields of Quantum Dreams Zenodo	10.5281/zenodo.10798144
038	2024	Presentation	MathArt Stream #8: William Blake and Active Inference Presentation	10.5281/zenodo.13711301
039	2024	Presentation	Sensemaking Federation: Exploring the Frontiers of Digital Innovation Presentation	10.5281/zenodo.14574046
040	2024	Paper	Shared Protentions in Multi-Agent Active Inference Entropy	10.3390/e26040303
041	2024	Paper	The Active Inference Institute & Active Inference Ecosystem (v2, 2024 snapshot) Zenodo	10.5281/zenodo.14108992
042	2024	Paper	Way Finding in the Infinite Imaginarium Zenodo	10.5281/zenodo.10601081
043	2024	Paper	Why Paleolithic Rockstars were both enigmatic and sporadic Physics of Life Reviews	10.1016/j.plrev.2024.04.010
044	2024	Paper	Writing on Curio Cards for the "On NFT" book Taschen	https://www.worldcat.org/isbn/978-3-8365-9970-2
045	2023	Paper	A guided tour through the spaces of particular "minds" Physics of Life Reviews	10.1016/j.plrev.2023.11.001
046	2023	Paper	A single-pheromone model accounts for empirical patterns of ant colony foraging Cognitive Systems Research	10.1016/j.cogsys.2023.02.005
047	2023	Paper	A snapshot and pipeline for tissue-specific gene expression meta-analysis in honey bees Zenodo	10.5281/zenodo.10400744
048	2023	Paper	A variational synthesis of evolutionary and developmental dynamics Entropy	10.3390/e25070964
049	2023	Paper	ATLAS: A Question Oriented Approach to Pattern Languages in Knowledge Management Zenodo	10.5281/zenodo.10296601
050	2023	Paper	An Account of Active Inference Modeling Zenodo	10.5281/zenodo.8415312
051	2023	Paper	Cognitive Sovereignty & Active Inference in the State of Exception Zenodo	10.5281/zenodo.10038231
052	2023	Paper	Comments on AI Accountability Policy to NTIA NTIA	10.5281/zenodo.8025956
053	2023	Paper	Distributed Science — The Scientific Process as Multi-Scale Active Inference OSF	10.31219/osf.io/dnw5k
054	2023	Paper	Enhanced NSF Postdoctoral Reporting via Synthetic Intelligence Language Processing Zenodo	10.5281/zenodo.10160656
055	2023	Paper	Experimental Entomology in the Age of Video JoVE	10.3791/65002
056	2023	Paper	Generalized Notation Notation for Active Inference Models Zenodo	10.5281/zenodo.7803327
057	2023	Paper	Generative Research Teams: Active Inference Compositions for Research and Meta-Science Zenodo	10.5281/zenodo.8164666
058	2023	Presentation	Of Ants & Aging Presentation	10.5281/zenodo.7855581
059	2023	Paper	Open Access science needs Open Science Sensemaking (OSSm) MetaArXiv	10.31222/osf.io/9nb3u
060	2023	Presentation	Postdoc review (2020–2023) Presentation	10.5281/zenodo.8377987
061	2023	Paper	The Active Inference Institute and Active Inference Ecosystem (v1) Zenodo	10.5281/zenodo.8266280



#	Year	Type	Title and Venue	Link
062	2023	Paper	The P3IF: Properties, Processes, and Perspectives Inter-Framework Zenodo	10.5281/zenodo.10034511
063	2023	Paper	To comment or not to comment Physics of Life Reviews	10.1016/j.plprev.2023.06.002
064	2023	Course	Transcript of Chris Fields, "Physics as Information Processing" course Zenodo	10.5281/zenodo.10447642
065	2023	Paper	William Blake & Buckminster Fuller: Lives in Juxtaposition Zenodo	10.5281/zenodo.7514367
066	2022	Paper	An Active Inference Ontology for Decentralized Science Zenodo	10.5281/zenodo.6320574
067	2022	Paper	Catechism for Towards Active Diffusion Zenodo	10.5281/zenodo.7443847
068	2022	Paper	From Users to (Sense)Makers: Stigmergic Social Annotation Hypertext '22	10.48550/arXiv.2205.06345
069	2022	Paper	On free will or the lack thereof (interview with Robert Sapolsky) ALIUS Bulletin	10.5281/zenodo.7394900
070	2022	Paper	Predictive Processing Interpretation of the Mirror Test Zenodo	10.5281/zenodo.7377255
071	2022	Book	Structuring the Information Commons: Open Standards and Cognitive Security COGSEC.org	https://cogsec.org
072	2022	Paper	Systems Modeling and Cognitive Audits for Hypercert Ecosystems Zenodo	10.5281/zenodo.7626768
073	2022	Paper	The Free Energy Principle & Active Inference: a Systematic Literature Analysis Zenodo	10.5281/zenodo.7449367
074	2022	Paper	TrustFinder: Recommendations for Community-Based Trust Systems Zenodo	10.5281/zenodo.7093836
075	2021	Paper	Active Inferants: An Active Inference Framework for Ant Colony Behavior Frontiers in Behavioral Neuroscience	10.3389/fnbeh.2021.647732
076	2021	Paper	Active Inference in Modeling Conflict Zenodo	10.5281/zenodo.5750934
077	2021	Paper	Digital Rhetorical Ecosystem Analysis: Sensemaking of Digital Memetic Discourse Zenodo	10.5281/zenodo.5573946
078	2021	Playbook	Innovator's Digital Playbook Web	https://innovatorsdigitalplaybook.com/
079	2021	Book	Narrative Information Ecosystems: Conflict and Trust on the Endless Frontier COGSEC.org	https://cogsec.org
080	2021	Paper	Narrative Information Management Zenodo	10.5281/zenodo.5565577
081	2021	Playbook	The Facilitator's Catechism Playbook Zenodo	10.5281/zenodo.4579413
082	2020	Paper	Active Inference & Behavior Engineering for Teams Zenodo	10.5281/zenodo.4021162
083	2020	Course	Communication for Remote Teams Udemy	https://www.udemy.com/course/communication-for-remote-teams/
084	2020	Paper	Distributed physiology and the molecular basis of social life in eusocial insects Hormones & Behavior	10.1016/j.yhbeh.2020.104757
085	2020	Paper	Emergent Teams for Complex Threats Zenodo	10.5281/zenodo.3986084
086	2020	Paper	Gene expression variation in the brains of harvester ant foragers is associated with collective behavior Communications Biology	10.1038/s42003-020-0813-8
087	2020	Course	How to use Keybase for remote teams Udemy	https://www.udemy.com/course/keybase-for-remote-teams/
088	2020	Paper	Measurement of natural variation of neurotransmitter tissue content in red harvester ant brains Analytical & Bioanalytical Chemistry	10.1007/s00216-019-02355-3
089	2020	Book	The Great Preset: Remote Teams and Operational Art COGSEC.org	https://cogsec.org
090	2020	Paper	The Innovator's Catechism Zenodo	10.5281/zenodo.4383229
091	2020	Paper	The neuroscience of decision making (interview with Timothy Hanks) ALIUS Bulletin	10.34700/8pp4-0h12
092	2019	Paper	Dennett Explained (interview with Daniel Dennett) ALIUS Bulletin	10.34700/7qkw-zh08
093	2019	Paper	PhD: Behavioral, Physiological, and Transcriptomic Variation Among Colonies of Pogonomyrmex barbatus Stanford University	http://purl.stanford.edu/pb813wm1484
094	2019	Paper	The ant colony as a test for scientific theories of consciousness Synthese	10.1007/s11229-019-02130-y



#	Year	Type	Title and Venue	Link
095	2019	Paper	The physiology of forager hydration and variation among harvester ant colonies Scientific Reports	10.1038/s41598-019-41586-3
096	2018	Paper	Foraging behavior and locomotion of the invasive Argentine ant from winter aggregations PLoS One	10.1371/journal.pone.0202117
097	2018	Presentation	MVEE: A Framework for Evolutionary Studies Presentation	10.5281/zenodo.13999298
098	2018	Paper	Of woodlice and men: A Bayesian account of cognition, life and consciousness (with Karl Friston) ALIUS Bulletin	10.34700/h460-nz89
099	2018	Paper	Partner Pen Play in Parallel (PPPiP): A New Paradigm for Relationship Improvement Arts	10.3390/arts7030039
100	2018	Paper	The Role of Dopamine in Collective Regulation of Foraging in Harvester Ants iScience	10.1016/j.isci.2018.09.001
101	2017	Paper	The MutAnts are here Cell	10.1016/j.cell.2017.07.046
102	2017	Paper	Two lineages that need each other Molecular Ecology	10.1111/mec.13964
103	2016	Paper	Ant Genetics: Reproductive Physiology, Worker Morphology, and Behavior Annual Review of Neuroscience	10.1146/annurev-neuro-070815-013927
104	2016	Paper	Context-dependent expression of the foraging gene in field colonies of ants Proceedings of the Royal Society B	10.1098/rspb.2016.0841
105	2016	Paper	Influence of Nuclear Structure on formation of radiation-induced lethal lesions Int. J. Radiation Biology	10.3109/09553002.2016.1144941
106	2015	Paper	Commentary: Portuguese crypto-Jews: the genetic heritage of a complex history Frontiers in Genetics	10.3389/fgene.2015.00261
107	2015	Paper	Could Ehrlichial Infection Cause Some Changes Associated with Leukemia? Medical Hypotheses	10.1016/j.mehv.2015.09.015
108	2015	Paper	Large Scale Coding Sequence Change Underlies the Evolution of Post-developmental Novelty in Honey Bees Molecular Biology & Evolution	10.1093/molbev/msu292
109	2026	Paper	Ento-Linguistics: Language, Ambiguity, and Scientific Communication in Entomology Zenodo	10.5281/zenodo.19574117
110	2026	Paper	A Living Meta-Analysis Architecture for Active Inference: Assertion Extraction, Nanopublications, and Hypothesis Scoring Active Inference Journal	10.5281/zenodo.19461933
111	2026	Paper	Dynamic Attentional Agents in Focused Attention Meditation: Hierarchical Computational Modeling of Expert-Novice Differences CSCIS vol 2857, Springer	10.1007/978-3-032-16955-6_11
112	2026	Paper	Compositional Approaches to Linguistic Case for Cognitive Modeling Active Inference Journal	10.5281/zenodo.19695259
113	2026	Paper	Towards Lean 4 Formalization of the Free Energy Principle: AI-Driven Theorem Sketching and Verification for Active Inference and Bayesian Mechanics Active Inference Journal	10.5281/zenodo.19699233
114	2020	Paper	The Facilitator's Catechism Zenodo	10.5281/zenodo.4062540
115	2026	Paper	The Architecture of False Gods: William Blake, Professor Jiang, and the Active Inference Corrective to Single Vision Zenodo	10.5281/zenodo.20144983
116	2026	Paper	Crescent City in Living Waves: Space, Time, People, and Minds on the Southern Cascadian Coast Zenodo	10.5281/zenodo.20286170
117	2026	Book	Introduction to Biology: A Generative Approach Zenodo	10.5281/zenodo.20286477
119	2026	Paper	Policy Entanglement in Active Inference Zenodo	10.5281/zenodo.20418904
120	2026	Paper	Bounded AutoResearch for a Tiny Reproducible Machine-Learning Task Zenodo	10.5281/zenodo.20417016
121	2026	Paper	Convergence Analysis of Gradient Descent Optimization Zenodo	10.5281/zenodo.20417136
122	2026	Paper	Editorial Quality at Scale: A Reproducible Prose-Review Pipeline Zenodo	10.5281/zenodo.20417104
123	2026	Paper	A template/ approach to Reproducible Generative Research Zenodo	10.5281/zenodo.20419007
124	2026	Paper	Active Inference Multi-Track Exemplar Zenodo	10.5281/zenodo.20417021



#	Year	Type	Title and Venue	Link
125	2026	Paper	BeeStack: An Evidence-Typed Scaffold for Whole-Colony Honeybee Simulation Zenodo	10.5281/zenodo.20420556
126	2026	Paper	When do bugs see (infra)red? Zenodo	10.5281/zenodo.20450880
127	2026	Paper	Self-Improvement Agent Harness: A Deterministic SIA Exemplar Zenodo	10.5281/zenodo.20453879
128	2025	Presentation	Biofirm Development with First Principles First and the Active Inference Institute at the Applied Active Inference Symposium 2024 Zenodo	10.5281/zenodo.14861596
129	2025	Paper	Con-cat-enate: emulation of Cat Hippocampus Zenodo	10.5281/zenodo.13626536
130	2025	Paper	Graphspeak: of language and handshake Zenodo	10.5281/zenodo.14737156
131	2025	Paper	BevCyc - posit on primitive drivers of creatures Zenodo	10.5281/zenodo.14737075
132	2025	Paper	Beacons - pre emulation of social cortex Zenodo	10.5281/zenodo.14737059
133	2025	Paper	Con-cat-enate: pilot overview Zenodo	10.5281/zenodo.14737042
134	2024	Paper	Push and Pull: A priming sequence Zenodo	10.5281/zenodo.10659374
135	2023	Paper	A Natural AI Based on The Science of Computational Physics, Biology and Neuroscience: Policy and Societal Significance Zenodo	10.5281/zenodo.10360056
136	2023	Paper	ATLAS: A Question Oriented Approach to the Use of Pattern Languages in Knowledge Management Zenodo	10.5281/zenodo.10362561
137	2023	Paper	Transcript of Active Inference GuestStream 049.1: "Clickbait, consciousness science, and responsible journalism" Zenodo	10.5281/zenodo.8228934
138	2023	Paper	Modern Nostr Index Card-based Knowledge Engineering Zenodo	10.5281/zenodo.8118155
139	2023	Paper	Slides for Iris Zenodo	10.5281/zenodo.7838652
140	2022	Paper	Canonical neural networks perform active inference Zenodo	10.5281/zenodo.7400525
141	2022	Paper	Interoception as modeling, allostasis as control Zenodo	10.5281/zenodo.7400708
142	2022	Paper	Transcript of: Mark Solms, "Consciousness as Precision Optimization: Some Physiological and Philosophical Considerations", ActInf GuestStream #016 Zenodo	10.5281/zenodo.7259006
143	2022	Paper	A Worked Example of the Bayesian Mechanics of Classical Objects Zenodo	10.5281/zenodo.7400785
144	2022	Paper	Transcript of discussions on: "Communication as Socially Extended Active Inference: An Ecological Approach to Communicative Behavior" Zenodo	10.5281/zenodo.7401874
145	2022	Paper	Tracking Public Sensemaking through Rhetorical Annotation of Image Memes Zenodo	10.5281/zenodo.6904426
146	2021	Paper	Transcript of: Karl Friston, 1st Applied Active Inference Symposium, Active Inference Lab, June 21, 2021 Zenodo	10.5281/zenodo.5797040
147	2021	Paper	Collaborative Writing for Catechism-Based Teams Zenodo	10.5281/zenodo.4624298
148	2020	Paper	Reimagining Maps Zenodo	10.5281/zenodo.4170025
149	2020	Paper	Infinite Games for Infinite Teams Zenodo	10.5281/zenodo.12601674
150	2026	Paper	The Music Never Stopped: A Grateful Data Compendium with a Category-Theoretic Interpretation Zenodo	10.5281/zenodo.20482025
151	2026	Paper	A Deterministic Testbed for Self-Organizing Agent-Team Coordination Zenodo	10.5281/zenodo.20533669
152	2026	Paper	Recovering LLM-Persona Accuracies from Unlabeled Votes Zenodo	10.5281/zenodo.20498699
153	2026	Paper	The Triplicate: A Data-Driven Large-Format Newspaper Layout Engine Zenodo	10.5281/zenodo.20533675
154	2026	Book	The Template Textbook Zenodo	10.5281/zenodo.20533125
155	2016	Paper	Lessons from the colony Science	https://www.science.org/doi/10.1126/science.352.6288.886
156	2016	Paper	Full speed ahead to the City on the Hill Science	10.1126/science.352.6288.886
157	2021	Paper	Disinforge Science	https://www.science.org/doi/10.1126/science.371.6524.22
158	2021	Paper	Defining events: 2020 in hindsight Science	10.1126/science.371.6524.22



#	Year	Type	Title and Venue	Link
159	2022	Paper	Active Blockference: cadCAD with Active Inference for Cognitive Systems Modeling International Workshop on Active Inference	10.1007/978-3-031-28719-0_10
160	2024	Paper	From Neuronal Packets to Thoughtseeds: A Hierarchical Model of Embodied Cognition in the Global Workspace arXiv	10.48550/arXiv.2408.15982
161	2021	Paper	Knowledge Management Archipelago Zenodo	10.5281/zenodo.5034808
162	2024	Paper	Comments Submitted by BlockScience, UW APL IRSIRI, and the Cognitive Security and Education Forum Zenodo	10.5281/zenodo.13273682
163	2017	Paper	Osteopathy and the Variable Variability of Health The Cranial Letter 70(4)	
164	2016	Paper	Cells, Mechanobiology, and Osteopathy The Cranial Letter 69(1)	
165	2026	Paper	iTrace: verification-first webcam eye-movement analysis Zenodo	10.5281/zenodo.20614908
166	2026	Paper	DemoCreate: Declarative Audio-Visual Demo Generation for Software Zenodo	10.5281/zenodo.20693216
167	2026	Paper	ENTO: an ENcrypted, Typed, Omnitrack container format for multimodal research data Zenodo	10.5281/zenodo.20396328
168	2026	Paper	GeneralizedNotationNotation (GNN) Zenodo	10.5281/zenodo.7803313
169	2026	Paper	COGANT: Deterministic Codebase-to-GNN Translation Zenodo	10.5281/zenodo.20705350
170	2026	Paper	AGEINT: Agentic Intelligence Zenodo	10.5281/zenodo.20732274
171	2026	Paper	On-Policy Distillation as Active Inference in Finite Variational Models Zenodo	10.5281/zenodo.20747834
172	2026	Paper	Template Madlib: Deterministic Token Injection for Conditional IMRAD Manuscripts Zenodo	10.5281/zenodo.20786638
173	2026	Paper	California Public Records: A Technical and Legal Reference for the Post-AB 473 Era Zenodo	10.5281/zenodo.20789899
174	2026	Paper	CogSecSkills: Multiharness Cognitive Security Skill Library Zenodo	10.5281/zenodo.20804585
175	2026	Paper	Realizing Emptiness: Operational Surrogates for No-Self-Evidence, QRF Opacification, and Bayesian Model Reduction Zenodo	10.5281/zenodo.20834846
176	2026	Paper	A Living Meta-Analysis of the Modafinil Literature Zenodo	10.5281/zenodo.20931964
177	2026	Paper	Refinement of Gold: A Metallurgical Analogy for Scientific Manuscript Composition Zenodo	10.5281/zenodo.20931955
178	2026	Paper	AlphaCOGANT: Recursive Corporate Self-Improvement as Active Inference Zenodo	10.5281/zenodo.20976824
179	2026	Paper	Mapping William Blake's Works: Evidence ledgers, source provenance, text-image diagnostics, and rights-bounded release controls Zenodo	10.5281/zenodo.21047573
180	2026	Paper	Sortition Upstream of NTQR Zenodo	10.5281/zenodo.21083779
181	2026	Paper	Exploratory Data Analysis: A Reproducible Notebook Template Zenodo	10.5281/zenodo.21086292
182	2026	Paper	A Domain Language for Specifying Controlled Methods Zenodo	10.5281/zenodo.21086548
183	2026	Paper	Entomological Law: A Field Map of Insects as Evidence, Threat, Property, Product, Patient, and Weapon Zenodo	10.5281/zenodo.21137276
184	2026	Paper	The Shape Between: A Full-Page Illustrated Storybook Template Zenodo	10.5281/zenodo.21176000
185	2026	Paper	Autopoietic Project Generation Zenodo	10.5281/zenodo.21227869



Software (95)

Repository	Stack	Description and Source Links
docxology/template https://github.com/docxology/template	Python	Production-grade scaffold for reproducible computational research — 10-stage DAG pipelines, ≥90% test coverage, cryptographic provenance, multi-project workspace, and agent-ready documentation · Zenodo software v3.3.0 · Zenodo paper: papers/2026_AutopoieticProjectGeneration/ zenodo: https://doi.org/10.5281/zenodo.20584820
docxology/QuadCraft https://github.com/docxology/QuadCraft	JavaScript	Minecraft variant built on tetrahedral geometry — voxels replaced by Quadray-coordinate tetrahedra, exploring Buckminster Fuller's synergetics in an interactive 3D world
docxology/MetaInformAnt https://github.com/docxology/MetaInformAnt	Python	Meta-framework integrating computational entomology, Active Inference, and information theory for modeling ant colony cognition and beyond paper: papers/2021_ActiveInferants/
docxology/codomymrnx https://github.com/docxology/codomymrnx	Python	AI-native modular coding workspace — 100+ modules, 500+ MCP tools, 21K+ zero-mock tests, multi-agent orchestration (Claude/Gemini/GPT), PAI integration
docxology/cognitive-engine https://github.com/docxology/cognitive-engine	Python	Hybrid neuro-symbolic AI combining fractal symbolic representations with probabilistic neural models for enhanced reasoning; includes Unipixel fractal building blocks and PEFf ethical reasoning
docxology/opentir https://github.com/docxology/opentir	Python	Comprehensive toolkit analyzing 250+ Palantir open-source repositories (6.2M+ lines of code) — automated discovery, multi-language analysis, dependency mapping, statistical profiling, and professional visualizations
docxology/timeline_generator https://github.com/docxology/timeline_generator	TypeScript	Networked Life Encoding — interactive D3 knowledge graph of human intellectual history; 24 relationship types, Perplexity AI enrichment; seeded around R. Buckminster Fuller's network
docxology/FORMINDEX https://github.com/docxology/FORMINDEX	Rich Text Format	Analysis and tooling around FORMIS, the world's largest ant-literature database; bibliometric mining and structured index generation
docxology/ActiveInferAnts https://github.com/docxology/ActiveInferAnts	Python	Active Inference for ants — multi-language implementations (Python) for modeling ant-colony cognition as Bayesian agents minimizing free energy paper: papers/2021_ActiveInferants/
docxology/crescent-city https://github.com/docxology/crescent-city	TypeScript	End-to-end pipeline for ingesting, parsing, and querying the Crescent City, CA municipal code; TypeScript-based legal-text processing
docxology/ivm-xyz https://github.com/docxology/ivm-xyz	Python	Python library for 3D and 4D Quadray/IVM (Isotropic Vector Matrix) geometry — coordinate conversion, volume calculations, and polyhedral operations
docxology/Markov_Blanket_Detection https://github.com/docxology/Markov_Blanket_Detection	Python	Dynamic Markov Blanket Detection (DMBD) — PyTorch-accelerated framework for identifying causal relationships in complex systems; static and temporal modes with cognitive-structure labeling
docxology/QuadMath https://github.com/docxology/QuadMath	TeX	Mathematical exposition of Quadray coordinate systems (4-vector tetrahedral coordinates); LaTeX documentation and computation library
docxology/active_inference https://github.com/docxology/active_inference	Python	Active Inference for, with, and by Generative AI — Python implementations of free-energy-minimizing agents integrated with modern AI tooling paper: papers/2020_BehaviorEngineering/
docxology/AgenticMesh https://github.com/docxology/AgenticMesh	Python	Active Inference & Agentic Mesh — modular Python infrastructure for building agentic systems using free-energy-minimizing Active Inference principles
docxology/curriculum https://github.com/docxology/curriculum	HTML	Curriculum development framework for interdisciplinary education in Active Inference, biology, and computational science
docxology/lean_niche https://github.com/docxology/lean_niche	Python	Lean/minimal niche-construction modeling — formal computational representations of ecological niche dynamics
docxology/literature https://github.com/docxology/literature	Python	Literature collection and analysis tools for structured ingestion, annotation, and exploration of scientific publications
docxology/obsidian-construction-from-text https://github.com/docxology/obsidian-construction-from-text	Python	Automated Obsidian vault construction from unstructured text — NLP pipelines to generate bidirectionally-linked knowledge graphs
docxology/qv_live_protocol https://github.com/docxology/qv_live_protocol	Python	QR code-based live protocol system for real-time data exchange and interactive session management



Repository	Stack	Description and Source Links
docxology/AGEINT https://github.com/docxology/AGEINT	HTML	Agentic Intelligence curriculum — modular educational scaffold for teaching bounded AI-agent support and synthetic analytic tradecraft · · Zenodo paper: papers/2026_AGEINT/ zenodo: https://doi.org/10.5281/zenodo.20732274
docxology/template_autoscientists https://github.com/docxology/template_autoscientists	Python	AutoScientists deterministic multi-agent scientific-discovery coordination harness · · Zenodo paper: papers/2026_DeterministicTestbedSelf/ zenodo: https://doi.org/10.5281/zenodo.20533669
docxology/template_newspaper https://github.com/docxology/template_newspaper	Python	The Triplicate data-driven large-format newspaper layout engine · · Zenodo paper: papers/2026_Triplicate/ zenodo: https://doi.org/10.5281/zenodo.20533675
docxology/ntqr_llm https://github.com/docxology/ntqr_llm	Python	Algebraic NTQR evaluation study recovering LLM-persona accuracies from unlabeled votes · · Zenodo paper: papers/2026_RecoveringLLMPersona/ zenodo: https://doi.org/10.5281/zenodo.20498699
docxology/itrace https://github.com/docxology/itrace	Python	Verification-first webcam eye-movement analysis toolkit for gaze, saccades, pupillometry, and diagnostic-quality exports · · Zenodo paper: papers/2026_ITrace/ zenodo: https://doi.org/10.5281/zenodo.20614908
docxology/coasys https://github.com/docxology/coasys	Python	Integration workspace aggregating and exercising the coasys (AD4M) agent-centric social-layer repositories
docxology/CogSecSkills https://github.com/docxology/CogSecSkills	Python	Multiharness cognitive-security and analytic-tradecraft skill library — 100 defensive SAT/cogsec skills runnable under Claude Code, Codex, and Hermes · · Zenodo paper: papers/2026_CogSecSkills/ zenodo: https://doi.org/10.5281/zenodo.20804585
docxology/template_madlib https://github.com/docxology/template_madlib	Python	Deterministic token-injection exemplar for conditional IMRAD manuscripts — reproducible madlib-style variable substitution for scientific writing · · Zenodo paper: papers/2026_TemplateMadlib/ zenodo: https://doi.org/10.5281/zenodo.20786638
docxology/ultralink-docx https://github.com/docxology/ultralink-docx	HTML	UltraLink document format tooling — linking, rendering, and transformation utilities for rich hyperlinked documents
docxology/cognitive https://github.com/docxology/cognitive	Python	Cognitive Ecosystem Modeling Framework — Active Inference agents with Obsidian-compatible knowledge management, bidirectional graph validation, belief updating, and network visualization
docxology/active_torchference https://github.com/docxology/active_torchference	Python	PyTorch-based Active Inference implementations — GPU-accelerated variational inference and free energy minimization
docxology/ant-pheromone https://github.com/docxology/ant-pheromone	Python	Computational simulation of ant pheromone trail dynamics and stigmergic communication
docxology/ant_stack https://github.com/docxology/ant_stack	Python	The Ant Stack — layered computational model of collective ant-colony intelligence paper: papers/2025_AntStack/
docxology/cohereants https://github.com/docxology/cohereants	Python	Computational research on insect infrared sensing and olfaction modeling · · Zenodo · Zenodo paper: papers/2026_WhenDoBugs/ zenodo: https://doi.org/10.5281/zenodo.20450970
docxology/ento_linguistics https://github.com/docxology/ento_linguistics	Python	Ento-Linguistics corpus pipeline — term extraction, terminology networks, semantic entropy, CACE scoring paper: papers/2026_EntoLinguistics/
docxology/cognitive_case_diagrams https://github.com/docxology/cognitive_case_diagrams	TeX	Compositional linguistic case for cognitive modeling — categorical case systems, DisCoCat/DisCoCirc diagrams, tests and figures for the Active Inference Journal manuscript paper: papers/2026_CognitiveCaseDiagrams/
docxology/Autonomous-Drone-Navigation-in-AirSim-With-Active-Inference https://github.com/docxology/Autonomous-Drone-Navigation-in-AirSim-With-Active-Inference	Julia	Active Inference-driven autonomous drone navigation in Microsoft AirSim simulation environment
docxology/biol-1 https://github.com/docxology/biol-1	HTML	BIOL-1: General Biology course materials — College of the Redwoods; lectures, labs, and interactive HTML content
docxology/biol-8 https://github.com/docxology/biol-8	HTML	BIOL-8: Human Biology course materials — College of the Redwoods; lectures, labs, and interactive HTML content



Repository	Stack	Description and Source Links
docxology/biology_textbook https://github.com/docxology/biology_textbook	Python	Open generative biology textbook — Markdown source, tested Python modules, programmatic figures; archived at Zenodo paper: papers/2026_BiologyTextbook/ zenodo: https://doi.org/10.5281/zenodo.20286478
docxology/cascadia https://github.com/docxology/cascadia	Python	Cascadia bioregion modeling and data analysis tools
docxology/course https://github.com/docxology/course	Python	Course materials framework for structured interdisciplinary learning modules
docxology/enactive_inference_model https://github.com/docxology/enactive_inference_model	Python	Three-level hierarchical enactive inference model of mental action — focused-attention meditation with expert/novice profiles; reproducible simulations and figures paper: papers/2025_Thoughtseeds/
docxology/EvoJump https://github.com/docxology/Evojump	Python	Comprehensive Framework for Evolutionary Ontogenetic Analysis — modeling developmental trajectories and evolutionary dynamics paper: papers/2025_EvoJump/
docxology/flick https://github.com/docxology/flick	Python	Lightweight Python tooling for rapid data flicking and processing pipelines
docxology/fuller-obsidian https://github.com/docxology/fuller-obsidian	unspecified	Obsidian vault dedicated to R. Buckminster Fuller's ideas — synergetics, geodesics, and design science
docxology/godel_ivm https://github.com/docxology/godel_ivm	Python	Gödel numbering meets the IVM (Isotropic Vector Matrix) — formal encoding of geometric objects as prime products
docxology/goference https://github.com/docxology/goference	Go	Active Goference — production-ready Go framework for Active Inference autonomous agents; genuine variational inference, free energy minimization, POMDP policy planning via gonum
docxology/hhs-opendata https://github.com/docxology/hhs-opendata	Python	Analysis of HHS (Department of Health and Human Services) open data — policy analytics and public-health data pipelines
docxology/grateful_data https://github.com/docxology/grateful_data	Python	The Music Never Stopped citation-bound Grateful Dead data compendium · · Zenodo · paper: papers/2026_MusicNeverStopped/ zenodo: https://doi.org/10.5281/zenodo.20482026
docxology/infra-calc https://github.com/docxology/infra-calc	JavaScript	Infrastructure calculator for estimating resource requirements and costs in multi-agent and AI systems
docxology/markdown_decision_processes https://github.com/docxology/markdown_decision_process	Python	Framework for Probabilistic Document Analysis — Markov Decision Process over Markdown artifacts for structured document reasoning paper: papers/2025_MarkdownDecisionProcess/
docxology/mdkv https://github.com/docxology/mdkv	Python	MKV-like key-value format and methods for Markdown — structured metadata embedding in plain-text documents paper: papers/2025_MDKV/
docxology/p3if https://github.com/docxology/p3if	Python	Properties, Processes, and Perspectives Inter-Framework (P3IF) — multiplexing interdisciplinary requirements frameworks to manage information risk and foster cognitive security paper: papers/2023_P3IF/
docxology/service https://github.com/docxology/service	Python	Service-layer infrastructure and API tooling for Python-based research applications
docxology/snake https://github.com/docxology/snake	Python	Classic snake game implemented in Python — used for testing multi-agent and reinforcement learning algorithms
docxology/steganographer https://github.com/docxology/steganographer	Rust	High-performance Rust tool for embedding BLAKE3+Ed25519 cryptographic signatures and visible watermarks into live video/audio via LSB steganography; four Rust crates, 132 tests, GStreamer integration
docxology/template_textbook https://github.com/docxology/template_textbook	HTML	Modular fillable scaffold for book-length technical works · · Zenodo paper: papers/2026_TemplateTextbook/ zenodo: https://doi.org/10.5281/zenodo.20533125
docxology/blake_jiang https://github.com/docxology/blake_jiang	Python	William Blake, Professor Jiang, and the Architecture of Intelligence — AI-augmented scholarly response to Game Theory #24 with Blake, Jiang, and Active Inference framing paper: papers/2026_BlakeJiang/



Repository	Stack	Description and Source Links
docxology/symergetics https://github.com/docxology/symergetics	Python	Symbolic Synergetics — computer algebra and formal symbolic systems for Buckminster Fuller's synergetics mathematics
docxology/transformer https://github.com/docxology/transformer	Python	Transformer architecture explorations — attention mechanisms and architectural variants for Active Inference research
ActiveInferenceInstitute/ActiveInferenceJournal https://github.com/ActiveInferenceInstitute/ActiveInferenceJournal	HTML	Content hub for the Active Inference Journal — 500+ video transcripts, stream archives, and multimedia publishing infrastructure for the Institute's primary educational output
ActiveInferenceInstitute/ActiveBlockference https://github.com/ActiveInferenceInstitute/ActiveBlockference	Jupyter	Active Inference agents in cadCAD block-based simulation — formal agent-based modeling using complex-systems simulation framework
ActiveInferenceInstitute/ActiveInferAnts https://github.com/ActiveInferenceInstitute/ActiveInferAnts	Python	Active Inference models for/of ants across 32+ programming languages — multi-language reference implementation of stigmergic Active Inference
ActiveInferenceInstitute/GeneralizedNotationNotation https://github.com/ActiveInferenceInstitute/GeneralizedNotationNotation	Python	GNN — formal notation standard for specifying Active Inference generative models; JSON/YAML schema, validators, and transpilers to Python, Julia, and MATLAB · Zenodo zenodo: https://doi.org/10.5281/zenodo.19600217
ActiveInferenceInstitute/fep_lean https://github.com/ActiveInferenceInstitute/fep_lean	Python	Lean 4 / Mathlib4 formalization of FEP-related mathematics — 50-topic sorry-free catalog, Hermes/OpenGauss LLM drafting, zero-mock verification, manuscript pipeline · · Zenodo paper: papers/2026_FEPLearn/ zenodo: https://doi.org/10.5281/zenodo.19699234
ActiveInferenceInstitute/cognitive https://github.com/ActiveInferenceInstitute/cognitive	Python	Cognitive modeling and simulation tools — multi-agent Active Inference environments, benchmarks, and visualization
ActiveInferenceInstitute/Active_Inference_Ontology https://github.com/ActiveInferenceInstitute/Active_Inference_Ontology	unspecified	Formal OWL/RDF ontology for Active Inference — defines classes, properties, and axioms for free energy principle concepts
ActiveInferenceInstitute/Journal-Utilities https://github.com/ActiveInferenceInstitute/Journal-Utilities	HTML	Transcription, processing, and publishing pipelines for the Active Inference Journal — Whisper-based download, transcription, RAG/knowledge-graph, and export utilities · Zenodo release zenodo: https://doi.org/10.5281/zenodo.18686966
ActiveInferenceInstitute/Symposium https://github.com/ActiveInferenceInstitute/Symposium	Python	Applied Active Inference Symposium materials — talks, workshops, proceedings, and collaborative session artifacts from annual Applied AIF symposia
ActiveInferenceInstitute/ActiveInferenceCategoryTheory https://github.com/ActiveInferenceInstitute/ActiveInferenceCategoryTheory	unspecified	Formal categorical foundations for Active Inference — string diagrams, functorial semantics, and compositional models using category theory
ActiveInferenceInstitute/CEREBRUM https://github.com/ActiveInferenceInstitute/CEREBRUM	Python	Case-Enabled Reasoning Engine with Bayesian Representations for Unified Modeling — grammatical case framework for compositional cognitive models · Zenodo paper: papers/2025_CEREBRUM/ zenodo: https://doi.org/10.5281/zenodo.15170907
ActiveInferenceInstitute/GEO-INFER https://github.com/ActiveInferenceInstitute/GEO-INFER	HTML	Geospatial Active Inference — applying free energy minimization to geographic systems, spatial planning, and bioregional intelligence
ActiveInferenceInstitute/AEOS https://github.com/ActiveInferenceInstitute/AEOS	unspecified	Active Entity Ontology for Science — OWL ontology for representing active entities, their actions, and causal relationships in scientific models
ActiveInferenceInstitute/courses https://github.com/ActiveInferenceInstitute/courses	HTML	Structured Active Inference learning courses — cohort-based curriculum materials, exercises, and collaborative learning artifacts
ActiveInferenceInstitute/Research-Discovery-Engine https://github.com/ActiveInferenceInstitute/Research-Discovery-Engine	HTML	AI-driven research synthesis and navigation — semantic search, citation graph analysis, and cross-domain connection discovery across the Active Inference literature



Repository	Stack	Description and Source Links
ActiveInferenceInstitute/Parr_et_al_2022_ActInf_Textbook https://github.com/ActiveInferenceInstitute/Parr_et_al_2022_ActInf_Textbook	Julia	Active Inference Textbook Study Group — code, exercises, and reading notes accompanying Parr, Pezzulo & Friston (2022)
ActiveInferenceInstitute/Biofirm https://github.com/ActiveInferenceInstitute/Biofirm	Python	Bioregional Active Inference for organizational stewardship — applying free energy minimization to ecological and community governance systems
ActiveInferenceInstitute/Start https://github.com/ActiveInferenceInstitute/Start	Python	Institute onboarding hub — getting-started guides, contributor pathways, project index, and orientation materials for new participants
ActiveInferenceInstitute/Knowledge-Engineering https://github.com/ActiveInferenceInstitute/Knowledge-Engineering	Jupyter	Knowledge engineering tools and workflows — ontology alignment, concept mapping, and structured knowledge-base construction for Active Inference
ActiveInferenceInstitute/ActInf_RxInfer https://github.com/ActiveInferenceInstitute/ActInf_RxInfer	unspecified	Active Inference with RxInfer.jl — Julia-based reactive message passing for efficient Bayesian inference in Active Inference models
ActiveInferenceInstitute/pymcp https://github.com/ActiveInferenceInstitute/pymcp	Python	PyMCP — Model Context Protocol bridge for PyMDP, enabling Active Inference agents to interoperate with MCP-compatible AI tooling
ActiveInferenceInstitute/GEN24 https://github.com/ActiveInferenceInstitute/GEN24	Jupyter	GEN-24 working project — collaborative generative systems research from the 2024 active inference cohort
ActiveInferenceInstitute/NoOrg https://github.com/ActiveInferenceInstitute/NoOrg	HTML	NoOrg is a Real Org — experimental organizational infrastructure and governance frameworks for decentralized scientific institutions
ActiveInferenceInstitute/ultralink https://github.com/ActiveInferenceInstitute/ultralink	unspecified	UltraLink — framework for linking, rendering, and transforming richly cross-referenced documents across multiple output formats
ActiveInferenceInstitute/ActiveInferenceImplementations https://github.com/ActiveInferenceInstitute/ActiveInferenceImplementations	Jupyter	Learning and exploring probabilistic inference — beginner-accessible Jupyter notebooks for Active Inference from first principles
ActiveInferenceInstitute/ATLAS https://github.com/ActiveInferenceInstitute/ATLAS	Python	Question-Oriented Pattern Languages for Knowledge Management — structured templates for navigating complex scientific knowledge spaces
ActiveInferenceInstitute/COGANT https://github.com/ActiveInferenceInstitute/COGANT	Python	Deterministic Codebase-to-GNN Translation — evidence compiler converting repositories into Generalized Notation Notation Active Inference artifacts via a 22-rule fixpoint engine; dual Python/Rust · · Zenodo · Zenodo paper: papers/2026_COGANT/ zenodo: https://doi.org/10.5281/zenodo.20705351
ActiveInferenceInstitute/on_policy_distillation https://github.com/ActiveInferenceInstitute/on_policy_distillation	Python	On-Policy Distillation and Active Inference — finite variational-model correspondence with deterministic witnesses, reproducibility artifacts, and paired publication release · · Zenodo · Zenodo paper: papers/2026_PolicyDistillationAs/ zenodo: https://doi.org/10.5281/zenodo.20749817
ActiveInferenceInstitute/Bots https://github.com/ActiveInferenceInstitute/Bots	unspecified	Institute bot infrastructure — automation, notification, and community management bots for the Active Inference Institute's platforms
ActiveInferenceInstitute/Garden-of-Iris https://github.com/ActiveInferenceInstitute/Garden-of-Iris	Python	Interactive visualization garden — generative art and data-visualization experiments exploring Active Inference dynamics
ActiveInferenceInstitute/gen25 https://github.com/ActiveInferenceInstitute/gen25	unspecified	GEN-25 working project — generative systems and collaborative research from the 2025 active inference cohort
ActiveInferenceInstitute/gnn https://github.com/ActiveInferenceInstitute/gnn	unspecified	Generalized Notation Notation v2 — next-generation GNN specification with extended model classes and improved toolchain
ActiveInferenceInstitute/ISSS https://github.com/ActiveInferenceInstitute/ISSS	Python	International Society for the Systems Sciences collaboration — joint research and educational materials bridging Active Inference and systems science
ActiveInferenceInstitute/topp https://github.com/ActiveInferenceInstitute/topp	TeX	The Open Problems Project — curated registry of open scientific problems in Active Inference and the free energy principle



Conferences, Posters, Seminars, Workshops (17)

2025 Inference about Foraging and Foraging as Inference | Queens College, CUNY, Biology

- Talk

Source: Queens College Biology <https://www.qc.cuny.edu/academics/bio/>

2025 Systems Processes, Active Inference, and Beyond | Enduring Patterns, Emerging Futures: Celebrating Dr. Len Troncale

- Talk 2025-09-17

Source: Zenodo presentation <https://doi.org/10.5281/zenodo.17138223>

2024 ...And Eternity in an hour of Systems Science Education and Knowledge Engineering | ISSS 2024

- Workshop

Source: ISSS <https://issis.org/world/>

2024 Funding AI public goods and keeping them public | Funding the Commons

- Panel

Source: Funding the Commons <https://fundingthecommons.io/>

2024 Intelligence in the Age of LLMs | European Identity and Cloud Conference 2024

- Panel

Source: KuppingerCole EIC speaker <https://www.kuppingercole.com/events/eic2024/speakers/3516>

2022 Tissue-specific gene expression meta-analysis in honey bees (Apis mellifera) | Entomology Society of America, Pacific Branch

- Talk 2022-04-10

Source: Entomological Society of America <https://entsoc.org/events>

2021 Thinking like a State: Active inference and the deep roots of complex societies | Cognition 2021

- Talk Avel Guenin-Carlut, Daniel Friedman, and Virginia Bleu Knight.

Source: Cognition <https://cognition.uqam.ca/>

2021 Introducing Active Inference as An Integrative Principle in The Study of Collective Intelligence | ACM-CI 2021

- Talk Avel Guenin-Carlut and Daniel Friedman.

Source: ACM Collective Intelligence <https://ci.acm.org/>

2020 Active Inference for Communicating Systems | IIBM, Chile

- Talk Invited online seminar. 2020-11-03

Source: IIBM <https://www.iibm.cl/>

2019 Invited seminar, Department of Animal Behavior, University of California, Davis | Essig Brunch Seminar Series

- Talk 2019-04-05

Source: UC Davis Entomology <https://entomology.ucdavis.edu/>

2018 Invited seminar, SUMS-RAS | Stanford University Mass Spectrometry Research Symposium

- Talk 2018-10-04

Source: Stanford Mass Spectrometry <https://mass-spec.stanford.edu/>

2018 Invited seminar, University of California, Berkeley | Essig Entomology Brunch

- Talk 2018-02-06

Source: Essig Museum <https://essig.berkeley.edu/>

2017 The role of dopamine in the regulation of foraging in the red harvester ant | Entomology 2017

- Poster

Source: Entomological Society of America <https://entsoc.org/events/annual-meeting>

2017 Quantitative LC-MS/MS analysis of dopamine in single ant brain samples | ASMS 2017

- Poster KM Krasinska, DA Friedman, DM Gordon, AS Chien.

Source: ASMS Annual Conference <https://www.asms.org/conferences/annual-conference>



- 2015 Transcriptomic analysis of the regulation of foraging in harvester ants | Biology and Genetics of Social Insects, CSHL 2015**
 - Poster DA Friedman, A Pilko, DM Gordon.
 Source: Cold Spring Harbor Meetings <https://meetings.cshl.edu/>
- 2014 Evolution of sex comb enhancers in the HOX gene Scr | Drosophila Research Conference 2014**
 - Poster DA Friedman, T Orenic, C McCullough, E Eksi, OY Barmina, A Kopp.
 Source: GSA Drosophila Conference <https://genetics-gsa.org/drosophila/>
- 2012 Visualizing the Integration of Sensory Neurons into a Sex-Specific Central Nervous System Circuit in 3-D | UC Davis Undergraduate Research Conference**
 - Talk DA Friedman.
 Source: UC Davis Undergraduate Research Center <https://urc.ucdavis.edu/>

Science Engagement, Outreach, Quotes, Articles (29)

- 2026 S3E10 The FEP: A Tool for Understanding and Modelling Complex Systems**
 - Video conversation on another channel
 Source: BathmikeC podcast <https://bathmikec.podbean.com/e/s3e10-the-fep-a-tool-for-understanding-and-modelling-complex-systems/>
- 2025 (DIA 1/SESSAO 1) Lancamento do Livro: As Colonias de Formigas Sao Conscientes?**
 - Video conversation on another channel
 Source: YouTube <https://www.youtube.com/watch?v=BJ-PAT3cgf4>
- 2024 From Ants to Active Inference with Daniel Ari Friedman**
 - Video conversation on another channel | Love and Philosophy, Andrea Hiott
 Source: YouTube https://www.youtube.com/watch?v=x_VPw1K55BM
- 2024 Building Smarter AI: Active Inference and Nested Models**
 - Video conversation on another channel | Bagaev, de Vries, Friedman, Pashea.
 Source: YouTube <https://youtu.be/u3w24FtAsKw>
- 2024 Interaction. In Search of Daniel Friedman's Deepest Value**
 - Video conversation on another channel | Math4Wisdom
 Source: YouTube <https://www.youtube.com/watch?v=T05uLRbcloE>
- 2023 Active Inference and The Free Energy Principle with Daniel Friedman**
 - Video conversation on another channel | No Way Out podcast
 Source: YouTube <https://www.youtube.com/watch?v=pYleeDb-B5E>
- 2023 Episode 6: Daniel Friedman - Active Inference Institute - What is Active Inference?**
 - Video conversation on another channel | Denise Holt
 Source: YouTube <https://www.youtube.com/watch?v=ZyFqRBp1KE0>
- 2022 Artist of Curio Cards 24-26 Shares Incredible Journey through Art and Science**
 - Video conversation on another channel | Jake Gallen
 Source: YouTube https://youtu.be/X_F2YFliUnY
- 2022 What Ant Behavior Reveals About Complexity Science with Daniel Friedman**
 - Video conversation on another channel | Making Sense of Complexity
 Source: YouTube <https://youtu.be/Or98-sG8r0> | YouTube episode <https://www.youtube.com/watch?v=4qsyV38qfTw>
- 2022 What is Active Inference?...with Daniel Friedman**
 - Video conversation on another channel | Inner Sense
 Source: YouTube <https://youtu.be/vHnm-zMqLvY>
- 2022 Great discussion with Daniel, OG Curio Card artist, cards 24, 25, 26**
 - Video conversation on another channel | ZeroG
 Source: YouTube <https://youtu.be/AHOGPsgIWmU>



2022 Daniel Friedman Interview - Adventures in NFTs #3

- Video conversation on another channel | World Crypto Network

Source: YouTube https://www.youtube.com/watch?v=mU_lwFsEpNg

2022 Banging the Drum with Daniel Friedman

- Video conversation on another channel | Nicholas Pacheco

Source: YouTube <https://www.youtube.com/watch?v=fBuS4yItitw>

2022 KB7 Fireside: Scaling Principled Games

- Video conversation on another channel | Kernel

Source: YouTube <https://www.youtube.com/watch?v=vFQk8UR-UVO>

2022 KBCast Episode 7: Kernel Block 5 Week 6 and special guest Daniel Ari Friedman

- Video conversation on another channel | Active Inference Lab

Source: YouTube <https://www.youtube.com/watch?v=R5zdTez8CPk>

2021 Daniel Friedman Interview - Adventures in NFTs #3

- Video conversation on another channel | World Crypto Network

Source: YouTube https://youtu.be/mU_lwFsEpNg

2020 Interview with Monica Kang about Complexity Weekend, with co-founder Shaun Swanson and organizer JP Gonzales

- Video conversation on another channel

Source: YouTube <https://youtu.be/nEaG4xi6MCE>

2021 Blurb in Science magazine inventing the word disinforme

- Blurb

Source: Science <https://www.science.org/doi/10.1126/science.abq0904>

2021 Complexity Weekend featured on InnovatorsBox blog, with quotes from Friedman and others

- Blurb

Source: InnovatorsBox <https://blog.innovatorsbox.com/complexity-weekend/>

2020 Blurb in Science magazine about rethinking time use for scientists

- Blurb

Source: Science <https://www.science.org/>

2016 Blurb in Science magazine about what ants can teach human city planners

- Blurb

Source: Science <https://www.science.org/doi/abs/10.1126/science.aag1520>

2013 Blurb in Science magazine about time travel

- Blurb

Source: Science <https://www.science.org/>

2017 Public lecture: Red Vic lecture series, San Francisco. History and philosophy of Art and Biology.

- Lecture

Source: verification <https://danielarifriedman.com/resume/verify.html>

2016 Public lecture: Red Vic lecture series, San Francisco. History and philosophy of self-organization, stigmergy, and eusocial insects.

- Lecture

Source: verification <https://danielarifriedman.com/resume/verify.html>

2021 Quoted in Dopamine Nation by Dr. Anna Lembke, p. 75: The world is sensory rich and causal poor.

- Quote

Source: Dopamine Nation <https://www.penguinrandomhouse.com/books/624957/dopamine-nation-by-anna-lembke-md/>

2025 No Signal 006: Daniel Ari Friedman and Jakub Smekal || Active Inference

- Profile or Interview

Source: No Signal <https://kernel0x.substack.com/p/no-signal-006-daniel-ari-friedman>



2022 Daniel Friedman: Drawing Decentralized Beauty From Biology

- Profile or Interview | Curio DAO

Source: Curio DAO Mirror https://mirror.xyz/0x8Fbf5c7Ca1295e892cC865500A8420C3316b889c/ztvis_d2IhlBUAb0LI4mCilrvomZr939O1T71a5-R1Y

2021 Written interview with RosyBVM

- Profile or Interview

Source: Rosy Conversation <https://rosybvm.com/2021/01/20/rosy-conversation-with-daniel-ari-friedman/>

2019 Radio Interview, KPOO 89.5 FM in San Francisco

- Radio

Source: KPOO <https://kpoa.com/>

Professional Participation, Engagement, and Service (27)

2022-2024 ISSS | Board of Directors

- Vice-President for Communication and Systems Education on the International Society for the Systems Sciences Board.

Source: ISSS <https://www.issss.org/home/>

2021-2023 JoVE | Academia

- Co-editor, Field and Laboratory Methods for Modern Entomology.

Source: JoVE methods collection <https://www.jove.com/methods-collections/567/field-and-laboratory-methods-for-modern-entomology>

2021 Cybernetics Society | Membership

- Member (MCyBS), Cybernetics Society.

Source: Cybernetics Society <http://cybsoc.org/>

2020-ongoing IPA | Membership

- Information Professional Association.

Source: IPA <https://information-professionals.org/>

2020-ongoing Active Inference Institute | Community

- Active Inference Institute co-founder.

Source: Active Inference Institute <https://www.activeinference.institute/>

2020-2023 Davis Complexity Group | Community

- Davis Complexity Group.

Source: verification <https://danielarifriedman.com/resume/verify.html>

2019-2024 Complexity Adventures | Community

- Complexity Adventures co-founder.

Source: Complexity Adventures <https://www.complexityadventures.com/>

2018-2021 ALIUS | Membership

- ALIUS Research Group contributor, editor, and co-organizer.

Source: ALIUS Research Group <https://www.aliusresearch.org/>

2018-2019 Wonderfest | Community

- Wonderfest Science Envoy; science outreach and community training.

Source: Wonderfest <https://wonderfest.org/>

2017-2021 ESA | Membership

- Entomological Society of America.

Source: Entomological Society of America <https://www.entsoc.org/>

2017-2019 Stanford | Academia

- NeuroChoice group, Stanford Neurosciences Institute.

Source: NeuroChoice <https://neurochoice.stanford.edu/purpose>

2017-2018 Stanford | Membership

- Graduate student member of Arts Committee for Stanford Biology department.

Source: Stanford Biology <https://biology.stanford.edu/>



2016-2019 Stanford Complexity Group | Community

- Stanford Complexity Group.

Source: Stanford <https://www.stanford.edu/> | verification <https://danielarifriedman.com/resume/verify.html>

2016 Stanford | Community

- Science Teaching Through Art (STAR) outreach program; poster presentations and conversations about ants at local middle and high schools.

Source: Stanford <https://www.stanford.edu/>

2015 Santa Fe Institute | Workshop

- Santa Fe Institute Complex Systems Summer School, one-month residential program.

Source: Santa Fe Institute CSSS <https://www.santafe.edu/engage/learn/programs/sfi-complex-systems-summer-school>

2014-2018 Gordon Lab | Field work

- Five summers of ant fieldwork during PhD under Prof. Deborah Gordon at Southwestern Research Station, Arizona.

Source: Southwestern Research Station <https://www.amnh.org/research/southwestern-research-station>

2014 Stanford | Field work

- Summer field ecology class with Stanford Prof. Rodolfo Dirzo at Las Tuxtlas tropical research station, Mexico.

Source: Los Tuxtlas Biological Station <https://www.ibiologia.unam.mx/eblt/>

2014 NIMBioS | Workshop

- Summer workshop on Evolutionary and Quantitative Genetics at NIMBioS, University of Tennessee, Knoxville.

Source: NIMBioS http://www.nimbios.org/tutorials/TT_eqq

2014 Science | Field work

- Drosophila work with Dr. Michael Lang in Arizona.

Source: Drosophila Evolution <https://droseu.net/drosophila-evolution/>

2014-ongoing Science | Academia

- Invited peer reviewer for 1-10 articles per year across different journals.

Source: Science <https://www.science.org/>

2014-2023 IUSSI | Membership

- International Union for the Study of Social Insects.

Source: IUSSI <https://ius.si.org/>

2014-2017 Stanford | Community

- Stanford Center for Evolutionary and Human Genomics outreach programs, including teaching genetics in local middle schools and hosting high school students at Stanford.

Source: Stanford Center for Evolution and Human Genomics <https://cehg.stanford.edu/>

2014-2017 Stanford | Community

- Stanford Bio Core Explorations hands-on lecture/workshop for undergraduates: Art and Biology.

Source: Stanford Biology <https://biology.stanford.edu/>

2013 UC Davis | Community

- Science mentoring at Woodland High School.

Source: Woodland High School <https://whs.wiusd.org/>

2012 UC Davis | Community

- Drums and percussion for UC Davis production of RENT.

Source: UC Davis <https://www.ucdavis.edu/>

2012-2013 UC Davis | Community

- Volunteer in emergency room at hospital in Sacramento, four hours per week for six months.

Source: UC Davis Health <https://health.ucdavis.edu/>

2011-2014 UC Davis | Community

- Chess Club at UC Davis President (2012-2014), organizing free outdoor chess at the farmer's market twice weekly for two years.

Source: UC Davis Campus Recreation <https://campusrecreation.ucdavis.edu/>



Art Portfolio and Uses (14)

All Art portfolio

- Posting drawings to an online portfolio since 2009.

Source: Flickr http://flickr.com/photos/daniel_friedman/

2021 Curio Cards

- A full Curio Cards set sold at Christie's on October 1, 2021 for 393 ETH, approximately \$1.2 million.

Source: Christie's Curio Cards lot <https://www.christies.com/lot/lot-6337619>

2021 Curio Cards

- Curio Cards collection stored in the Arctic World Archive with art and provenance data intended for long-duration preservation.

Source: Top Dog Beach Club <https://topdogbeachclub.com/non-fungible-vault/>

2021 Curio Cards

- Public auction and market history for Curio Cards includes major 2021 sales; verify venue-specific claims before reusing exact auction-house wording.

Source: Sotheby's Curio Cards <https://metaverse.sothebys.com/natively-digital/lots/curio-cards-complete-set-1-30-plus-17b>

2019 Art used

- Art used in the eBook Rebellng with Care: Exploring open technologies for commoning healthcare.

Source: ResearchGate https://www.researchgate.net/publication/335911369_Rebellng_with_Care_Exploring_open_technologies_for_commoning_healthcare

2019 Art used

- Art used in Slate.fr blog.

Source: Slate France <https://www.slate.fr/story/173103/fecondation-spermatozoide-ovule-chevalier-belle-au-bois-dormant-mythe-sexisme-naturalisation-biologie?>

2019 Art used

- Art used in degrowth blog.

Source: Degrowth.info <https://www.degrowth.info/blog/on-strategies-for-socioecological-transformation>

2018 Art used

- Art used in Medium post by Lorena Cabeza, August 23, 2018.

Source: Medium <https://medium.com/@lorenacabeza/manifiesto-la-escritura-y-la-vida-10d4c254c0bf>

2018 Art used

- Cover image of Probe magazine.

Source: Probe Magazine <https://www.ssbprobe.com/decentraldogma-danielfriedman>

2017 Curio Cards

- Early Ethereum art NFT project; art for Curio Cards 24, 25, and 26.

Source: Curio Cards artist <https://curio.cards/artist/danielfriedman/> | Curio Cards docs <https://docs.curio.cards/the-artists/daniel-friedman>

2017 Art used

- Art used in Bas Haring blog.

Source: Me Judice <https://www.mejudice.nl/artikelen/detail/mijn-economiemuseum-de-markt>

2017 Art used

- Art used in Vice Motherboard blog.

Source: VICE Motherboard <https://www.vice.com/en/article/ipagk8/a-dollar25-million-book-heist-proves-the-printed-page-is-still-sacred>

2016 Art used

- Featured in Genetics Society of America art blog.

Source: Genes to Genomes <https://genestogenomes.org/gsa-art-daniel-friedman/>

2016 Art used

- Art featured by Stanford Art of Neuroscience competition.

Source: Stanford Scope <https://scopeblog.stanford.edu/2016/07/06/art-of-neuroscience-competition-highlights-beauty-of-the-brain/>

